

Designing the "Front": An Overview of Profile Elements on Social Network Sites

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ABSTRACT

User profile is one of the most fundamental approaches on Social Network Sites (SNS) to make an impression on others. Designing user profiles is a process of designing users' personal "front" [6, 14]. This study investigates profile elements across 110 SNS and identifies 62 elements with two groups of attributes: manually filled by users or automatically generated by the system, providing individual or social information. We annotated the presence of elements and whether they are optional or concealable. The most common elements are Profile Image, Bio Text, Username, Location, and Link to Other Social Media. We found SNS' proportion of system-generated elements is positively correlated with social elements. SNS can be categorized into "Real Life Mirror" and "Showcase as Need" based on the number of profile elements and can be categorized into "Resume" and "Convenient Social" by the attributes of elements. We discuss the potential use of the data and the implications for future research. This work contributes to CSCW community by offering a snapshot of update-to-date profile design with comprehensive analysis.

CCS CONCEPTS

• Human-centered computing → Empirical studies in HCI.

KEYWORDS

social network sites, SNS, profile, design

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1 INTRODUCTION AND BACKGROUND

According to Goffman's impression management theory, humans are always performing for others, trying to make a particular impression [14]. It happens not only in face-to-face (F2F) but also in online communication. User profile is an integral part of Social Network Sites (SNS) for self-disclosure and self-presentation [6, 20]. Scholars reported that people strategically manipulate information on profiles [11, 24, 25] and the methods vary across cultures [30].

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Many studies have investigated how profile elements are relevant to attitudes, behaviors, and social network [16, 20, 22]. People who provide a username are more involved in online sharing and gain more recognition from peers than anonymous users [22]. For freelancers, having no profile image has a negative correlation, while having a non-human profile has a positive correlation with the number of reviews [16]. Lampe et al. [20]'s study of Facebook profile elements in the early 2000s revealed that users who populated a field tend to have more friends than those who did not populate the same field. Cress et al. [10] found profiles with off-topic information are less preferred by members in common identity communities, but not members in common bond communities. The amount of information presented on the profile may affect members' liking or ratings towards others [3, 13].

Although SNS users have much freedom of editing their profiles, what elements are available and if the fields are optional or concealable is determined by site designers. For example, Facebook launched a real name policy requiring users to provide "the name they use in real life" [15] while some sites such as 4chan even do not require an account to post content. The design of profile elements is a process of "engineering a new personal front" [6], which indicates the founders and designers' values, and can also shape community members' perception and judgement of others.

Despite a huge body of research has investigated individual profile elements, little work has systematically reviewed them across various SNS. This study aims to offer a taxonomy of the profile elements of different types of SNS and online communities and aims to serve as a snapshot of state-of-the-art SNS profile design practices. The following research questions are asked: (1) What existing design elements are presented on the state-of-art SNS profiles? (2) How can SNS be categorized based on the profile elements?

2 METHODS

We narrow down the original definition of SNS [5] and refer to them as web-based services that allow individuals to (1) construct a public or semi-public profile within a bounded system, (2) articulate a list of users with whom they share a connection, (3) post content to interact or self-present. Instant messaging and multiplayer games may have the above feature but their primary services are not for self-presentation or public connection building. Dating apps as a special kind of SNS provide one-one matching instead of general social connections. These sites are not included in this study.

SNS Selection. An initial list with 271 SNS was retrieved from multiple sources [28, 29]. SNS were selected based on the following criteria. 1) actively serve as a general-use SNS, 2) allow new user registration in the US (the author's location), 3) offer free services, 4) represent as popular SNS in the regions they originated. The number of claimed user registration, such as more than one million,

Element	Description	Source	Info Type	Coverage
User Name	Pseudonym a user created to identify themself. Usually a username has to be unique	User	Individual	75.45%
Real Name	The site requests a user to provide "real name" or a name in the format of first name last name	User	Individual	40.91%
Bio-text	A text field where users can free-type anything about themselves	User	Individual	89.09%
Birthday / Astrological Sign	Month or day of birthday, or astrological sign derived from birthday	User	Individual	24.55 %
Employment	Including job title, employer, industry, or employment history	User	Individual	35.45%
Organization	Organizations the user joins in real life, e.g., student unions, professional associations	User	Social	4.55%
Virtual Points	Numeric value of a user's reputation or experience points, e.g., karma	System	Individual	15.45%
Virtual Achievements	Categorical achievements that a user obtain after completing certain tasks or reaching certain points, e.g., level 4, gold trophy, "500 comments"	System	Individual	25.45%
# Followers	How many people follow this user. Following does not need another user's approval	System	Social	57.27 %
List of Followers	Who follows the user	System	Social	50.91 %
# of Views	How many times a user's profile or posts are viewed by others	System	Social	25.45%
# of Social Interactions	How many times others interact with the user, e.g., be liked, be saved, receive messages	System	Social	36.36%
Stats of Activities	e.g., total # of posts, # of posts today, avg ratings	System	Individual	55.45%
Online Status	If a user is online, or the last active time	System	Individual	24.55%

Table 1: Examples of Profile Elements

can be seen as an indicator as popularity. Most SNS are from the US, but we tried to include SNS in diverse regions and languages. The filtering left us with 110 SNS originating from 23 countries. The foundation year of selected sites ranged from 1995 to 2021. The average history was 14.38 years.

Coding Schema. A list of profile elements was initialized from the literature [6, 20], including Username, Age, Gender, About me, Interests, and Political Views. We expanded the element list during the annotation. After several rounds of iteration, we developed a coding schema with 62 profile elements. These elements can be categorized by source (created by user or system) and information type (individual or social). An element is categorized as individual if it conveys information about the user themselves or it helps to identify the user, such as username, birthday, and employment. If categorized as social, the element indicates the user's social activities, such as List of Friends and Communities. Example elements are listed in Table 1.

All the elements were annotated with three values: 0 (*not exist*), 1 (*optional or concealable*), and 2 (*required or public*). Optional elements are those that do not affect the basic functions of the service if a user did not fill the fields, such as Profile Image. Concealable elements mean users can hide or restrict access to the fields, such as Friend List and Online Status. Some elements may not be concealable, such as Date Joined, Virtual Points, and Virtual Achievements, which are designed to be public. Fields that are required during registration, such as Date of Birth and Phone Number, but does not show on the profile were coded as 0.

Annotation Procedure and Data Analysis. We created an account and annotated the elements on each site. For ambiguous fields, such as if a friend list or badge is concealable, we referred to the sites' official documentation and user forums. We also observe public profiles in case some elements are not available for new users. For sites that request a valid phone number or request identity verification, personal accounts were used as the data source. Google Translate was applied on sites that do not have an English version. The annotation iterated for several rounds as new profile elements were introduced. Only the design elements on profiles instead of users' information were annotated.

The data reported in this study were collected in early 2022. Data collection was completed on a spreadsheet and data analysis was completed with Python. The annotated data is available here: <https://osf.io/29swz/>.

3 FINDINGS

We first report the proportion and attributes of the 62 profile elements (RQ1). Next we selectively present the results of some interesting individual elements. Then we categorize SNS into four types by the design of profile elements (RQ2).

Profile Elements and their Attributes. Most elements are defined by users (54.84%) and are individual information (66.13%). Among self-defined elements, 94.12% are individual information. Among social elements, 90.48% are automatically generated by the system. Details are in Figure 1. A site's proportion of social elements

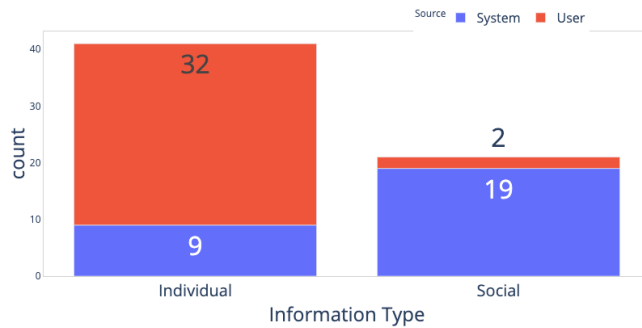


Figure 1: Profile elements by attributes

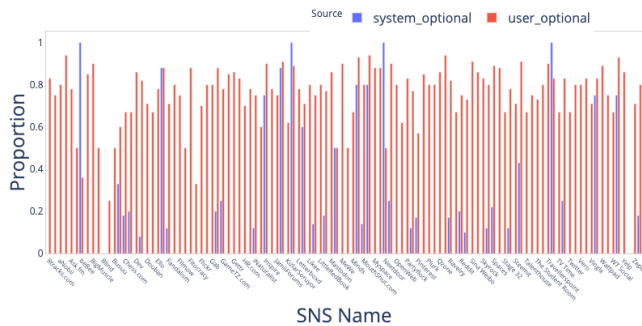


Figure 2: % of Optional elements created by user and system

is positively correlated with the proportion of system-generated elements ($r(108) = 0.71, p < 0.01$), as shown in Figure 5.

Descriptive Statistics of Individual Elements. The most common elements are Profile Image (95.45%), Bio Text (89.09%), Username (75.45%), Location (75.45%), and Link to Other Social Media (58.18%). **Gender** is one of the most studied online identities [4, 17].

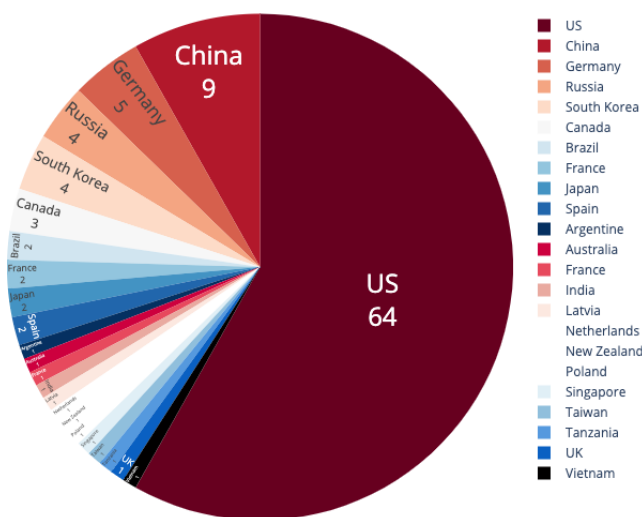


Figure 3: Number of SNS by country of origin

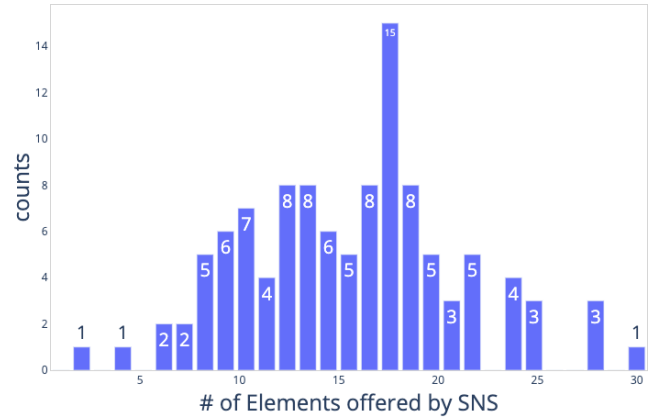


Figure 4: Distribution of number of profile elements

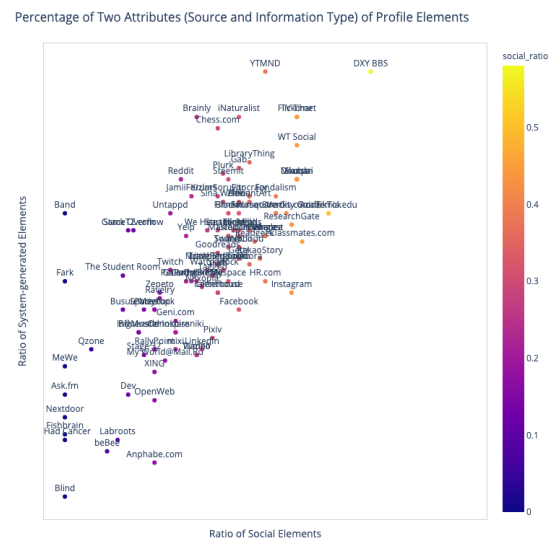


Figure 5: Percentage of Source (Created by User / System) and Information Type (Individual / Social) of Profile Elements across 110 SNS

Though many platforms ask for gender information in the registration, this information may not be public on the profile. Only 31.82% of SNS have a field for gender and 9.09% of SNS have a field for pronouns. As mentioned in the literature, **Age** is hardly detectable online [6]. Some SNS (34.55%) offer a field to show their age or year of birth, while 24.55% give the options of month and day only, or astrological sign. Though it is rarely scientifically tested, many people believe that personality is relevant to the sun sign [27], which may be a reference for future interaction. The use of game mechanics such as points and badges has been applied to SNS for many years to encourage engagement and contribution [12, 26]. Our data show that 30.91% of SNS, such as Stack Overflow, Brainly, Chess.com, and Reddit, display **Virtual Points** or **Virtual Achievements** on user

profiles. These sites rely on users' contribution to make the community sustainable. Some elements are used as social gamification metrics [18]. The most common social metrics are *#of Followers* (57.27%), *# of Friends* (32.73%), *# of Social Interactions* (36.36%), and *# of Views* (25.45%). Most sites display these fields by default, fewer than 10 sites let users hide these social metrics.

Categories of SNS by Profile Elements. On average each site has around 15.07 profile elements. SNS with the most number of elements are Geni.com (29), LinkedIn (27), Nexopia (27), Spaces (27), Facebook (24), and LiveJournal (24). The distribution of the number of profile elements per site is in Figure 4. Based on the number of profile elements used, SNS can be categorized into two types. *Real Life Mirror* SNS tend to include as many fields as possible, and they try to "duplicate" a user on the Internet. *Showcase as Needed* SNS only display a certain amount of information that is considered sufficient for interaction on the site. If users want to disclose more, they can write in the free text "About Me" field. These two types point to the left and right end of Figure 4.

SNS can also be categorized by the proportion of elements source and information type. On *Resume* SNS, people usually need to spend lots of effort manually editing their resume, and most information is about themselves. This type points towards the bottom left corner of Figure 5. *Convenient Social* SNS on the top right corner of Figure 5 asks for little manual effort to manage the profile and helps users present social information. The breakdown of element source and information type is shown in Figure 1.

Interestingly, we also found profile elements' visibility is associated with information source. Visibility can be understood as user control, indicating if the user can hide or choose not to disclose certain information. As described in Section 2, optional or concealable elements were coded as 1, required or public elements were coded as 2. In general, self-defined fields have more control than system-generated fields. A Wilcoxon signed-rank test showed that the proportion of a site's optional self-defined elements over all of its self-defined elements ($M = 74.65\%$) was higher than the proportion of a site's concealable system-generated elements over all of its system-generated elements ($M = 13.74\%$, $p < 0.01$). Thirty-six sites have the mechanism of adding friends while 12 of them allow users to hide or restrict access to their friend list. Four out of 27 SNS allow users to hide online status. We can evaluate a site's degree of user control based on the proportion of optional elements (Figure 2).

4 DISCUSSION AND CONCLUSION

SNS have evolved for many years since its emergence, and elements in SNS profiles have huge changes compared with those decades ago [2, 20, 23]. As SNS keep updating, profile elements are becoming increasingly diverse.

Implications for Future Work. We provide the report and annotated data (<https://osf.io/29swz/>) for SNS designers and researchers to learn the state-of-art SNS profile design. For example, designers of a new SNS can refer to similar services and decide which type of SNS they want to build, whether it will be close to a Resume or Convenient Social site. Researchers can use this work as a snapshot and compare it with past or future SNS design trends.

Besides that, we propose two potential research directions. The first is the effects of similarity information on social impression and intention to connect. For example, Twitter and Instagram show if someone is followed by people a user is following [1]; LinkedIn and Facebook show mutual connections [21]; LibraryThing lists common communities; Readgeek and Zhihu display books and questions in common. Common connections can help users identify which community the person is in. Things in common can help people establish common ground [9] and generate a positive impression [7], which may benefit future interactions. Some sites display compatibility or similarity values calculated by their own algorithms, such as Filmnow and ReadGeek. It is interesting to see if common connections, communities, things, or similarity scores have different effects on people's impression of others and their intention to connect.

Another direction is the involvement of AI. For example, future profiles may be created with the assistance of AI [8, 19]. It is to study users' perceptions of their own profiles created by AI. How will people react to different types of profile elements selected by AI? How does it affect people's intention to connect or make friends?

Limitation. This work has the following limitations. First, the sample of 110 sites does not include all the SNS and may exclude underrepresented sites. Future work can expand the data source to more diverse online communities. Second, most annotations used new accounts as the data source, which may introduce biases or inaccuracy.

Conclusion. This study annotated 62 profile elements across 110 SNS and analyzed the elements by their source and information type. We categorize SNS into "Real Life Mirror" and "Showcase as Need" based on the number of profile elements; SNS can also be categorized into "Resume" and "Convenient Social" by the attributes of elements. We found most social elements are generated by the system, and users have less control over system-generated elements than self-defined elements. This work offers a snapshot of update-to-date profile design with comprehensive analysis.

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